

N+- Network Fundamentals and building network

[Q] Firewall Basics Theory (1 hour): Firewall types and use cases (hardware and software firewalls). Practical (1 hour): Configure basic rules in a software firewall (e.g., Windows Defender). Task: Document how to block and allow specific traffic using a firewall.

[Q-1] Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

[ANS]

○ Steps to Create an Outbound Rule for Spotify

1. Open **Control Panel**.
2. Click **Windows Defender Firewall**.
3. Select **Advanced Settings**.
4. In the left pane, click **Outbound Rules**.
5. In the right pane, click **New Rule**.
6. Select **Program** and click **Next**.
7. Choose **This Program Path** and browse to the Spotify executable file (Spotify.exe).
8. Click **Next**.
9. Select **Block the Connection** and click **Next**.
10. Select all profiles (**Domain, Private, Public**) and click **Next**.
11. Enter the rule name **Block Spotify Traffic**.
12. Click **Finish**.

[Q-2] Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

[ANS]

○ Steps to Create an Inbound Rule

1. Open **Windows Defender Firewall with Advanced Security**.
2. Click **Inbound Rules**.
3. Select **New Rule**.
4. Choose **Program** and click **Next**.
5. Browse and select **chrome.exe**.
6. Click **Next**.
7. Select **Allow the Connection**.
8. Click **Next**.
9. Select the required network profiles (**Domain, Private, Public**).
10. Click **Next**.
11. Open the rule properties after creation.
12. Go to the **Protocols and Ports** tab.
13. Select **TCP**.
14. Specify **Local Port 443**.
15. Click **Apply** and then **OK**.

[Q-3] Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall. Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.

[ANS]

○ **Steps to Block a Specific IP Address**

1. Open **Windows Defender Firewall with Advanced Security**.
2. Click **Inbound Rules**.
3. Select **New Rule**.
4. Choose **Custom Rule** and click **Next**.
5. Select **All Programs**.
6. Click **Next** until you reach the **Scope** tab.
7. Under **Remote IP Address**, select **These IP Addresses**.
8. Click **Add**.
9. Enter the IP address:
123.45.67.89
10. Click **OK**.
11. Click **Next**.
12. Select **Block the Connection**.
13. Click **Next**.
14. Select all profiles (**Domain, Private, Public**).
15. Click **Next**.
16. Enter a rule name such as **Block IP 123.45.67.89**.
17. Click **Finish**.

[Q-4] Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

[ANS]

Feature	Hardware Firewall	Software Firewall
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Installation	Dedicated physical device	Installed on a computer or server
Protection Scope	Protects the entire network	Protects a single device
Performance Impact	Minimal impact on client devices	Uses local CPU and memory resources

○ Hardware Firewall Advantages

1. Protects multiple devices simultaneously.
2. Centralized security management.
3. Suitable for large organizations and offices.

○ Use Case

A college or company network uses a hardware firewall to protect hundreds of computers connected to the internet.

○ Software Firewall Advantages

1. Easy to install and configure.
2. Provides protection for individual devices.
3. Can create application-specific rules.

○ Use Case

Windows Defender Firewall protects a personal laptop by controlling incoming and outgoing network traffic.